

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2 hours 30 minutes

Version: 1

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

1. **C — Measuring tape** 1 mark

A **metre rule** is normally about 1 m long and a **vernier caliper/screw gauge** measure only very small lengths (a few cm or mm), so none can span 2 m in one reading. A **measuring tape** is a flexible ruler made for larger distances (several metres), so it is the practical choice here. A physical balance measures mass, not length.

2. **B — Systematic error** 1 mark

A **systematic error** is a consistent error, of the same size and direction in every reading, usually caused by a fault in the instrument (e.g. wrong zero/calibration). Since every reading here is 2 g too high in the same direction, this is systematic, not random (random errors scatter unpredictably above and below the true value).

3. **B — Vibratory motion** 1 mark

A pendulum bob moves **to and fro about a mean (equilibrium) position** — this repeated back-and-forth motion is called **vibratory (oscillatory) motion**. Rotatory motion is spinning about an axis; translatory motion is movement from one point to another; random motion has no fixed pattern.

4. **A — 4.5 m/s²** 1 mark

Deceleration $a = \Delta v/t = (18 - 0)/4 = 4.5 \text{ m/s}^2$. The bus loses 18 m/s of speed in 4 s, so it slows down at a uniform rate of 4.5 m/s². (Deceleration is negative acceleration; only the magnitude is asked here.)

5. **C — Newton's third law** 1 mark

Newton's **third law** states that for every action there is an equal and opposite reaction. The rocket pushes hot gas downward (action); the gas pushes the rocket upward with equal force (reaction), propelling it forward. This is a classic example of action-reaction force pairs, not $F = ma$ directly (that describes the size of acceleration, not the pairing of forces).

6. **A — 60 kg·m/s** 1 mark

Momentum $p = mv = 15 \text{ kg} \times 4 \text{ m/s} = 60 \text{ kg·m/s}$. Option B (3.75) results from dividing instead of multiplying; option C (19) from wrongly adding mass and velocity; option D (30) from a halving error.

7. **B — Like parallel forces** 1 mark

When two forces act **parallel to each other in the same direction**, they are called **like parallel forces** — exactly as when two people push a cart from behind in the same direction. If the forces were parallel but opposite in direction (e.g. turning a bike's handlebar), they would be **unlike parallel forces**.

8. **A — 27 N·m** 1 mark

Torque $\tau = F \times d = 45 \text{ N} \times 0.6 \text{ m} = 27 \text{ N·m}$. Option D (45.6) comes from wrongly adding F and d instead of multiplying; option C (0.013) from dividing instead of multiplying.

9. **A — 400 N/m** 1 mark

By Hooke's law, spring constant $k = F/x = 40 \text{ N} / 0.10 \text{ m} = 400 \text{ N/m}$ (extension converted from 10 cm to 0.10 m before dividing). Option D (4000) is a common error from forgetting to convert cm to m.

10. **A — 5000 Pa** 1 mark

Pressure $P = F/A = 150 \text{ N} / 0.03 \text{ m}^2 = 5000 \text{ Pa}$. Option D (50,000) results from misplacing a decimal point in the area; option B (500) from a factor-of-10 slip.

11. B — Tidal 1 mark

As the Moon orbits the Earth, its gravitational pull (along with the Sun's, to a lesser extent) causes bulges in the ocean, known as **tides**, which occur twice a day. **Tidal energy** harnesses this rise and fall of sea level (tidal range) or tidal currents to generate electricity, e.g. using a tidal barrage across a basin. Geothermal comes from the Earth's internal heat; biomass from organic matter; wave energy from surface wind-driven waves, not the Moon's pull.

12. D — Unfalsifiable 1 mark

A good scientific statement must be **falsifiable** — it must be possible, at least in principle, to design an experiment or observation that could prove it wrong. A statement that can *never* be tested or disproved by any conceivable experiment is called **unfalsifiable**, and falls outside the scope of science (regardless of whether it happens to be true).

SECTION B — Short Questions (11 × 3 = 33 Marks)**Q.2(i) — Main Question** 3 marks

What instrument would you use to measure a 12 m school corridor, and why is it better suited than a metre rule? State one precaution against parallax error.

Instrument: A **measuring tape** should be used. 1 mark

Reason: A metre rule is only about 1 m long, so measuring 12 m with it would require repositioning it many times, introducing cumulative error. A measuring tape is a flexible ribbon that can be pulled out to several metres in one continuous length, making the measurement faster and more accurate. 1 mark

Precaution against parallax error: The eye must be positioned directly (perpendicularly) above/in line with the scale marking being read, not at an angle, otherwise the reading will appear shifted. 1 mark

OR

Define least count. What is the least count of a metre rule graduated in millimetres? Is a reading of 1.234 m from it realistic? State how many decimal places she should report.

Least count: The least count of an instrument is the smallest measurement that can be accurately read from its scale. 1 mark

Least count of the metre rule: Since the smallest division is 1 mm, its least count is **1 mm (0.001 m)**. 1 mark

Is 1.234 m realistic? Yes — since the least count is 0.001 m (1 mm), a reading to **three decimal places** (the thousandths, i.e. the nearest mm) is exactly consistent with what the rule can resolve, so 1.234 m is a justified reading. She should *not*, however, report a fourth decimal place (e.g. 1.2340 m), as that would falsely imply a precision finer than the rule's least count. 1 mark

Q.2(ii) — Main Question 3 marks

Compare a vernier caliper and a screw gauge: which has the smaller least count, and what type of object is each best suited to measure?

Smaller least count: A **screw gauge** generally has a smaller least count (typically 0.01 mm) than a **vernier caliper** (typically 0.01–0.1 mm depending on the number of vernier divisions), making the screw gauge more precise for very fine measurements. 1 mark

Vernier caliper: Best suited for measuring the external/internal dimensions and depth of moderately small objects, e.g. the diameter of a marble or the thickness of a coin. 1 mark

Screw gauge: Best suited for measuring very small thicknesses or diameters, e.g. the diameter of a thin wire or the thickness of a sheet of paper. 1 mark

OR

A screw gauge has pitch 0.5 mm and 25 divisions. Calculate its least count. If the main scale reads 2 mm and the 15th circular division coincides with the datum line, find the wire's diameter.

Given: Pitch = 0.5 mm, circular scale divisions = 25 **Required:** Least count, diameter

Least count = Pitch / No. of circular divisions = 0.5 mm / 25 = **0.02 mm** 1 mark

Circular scale reading = 15 × 0.02 mm = 0.3 mm 1 mark

Diameter = Main scale reading + Circular scale reading = 2 mm + 0.3 mm = **2.3 mm** 1 mark

Q.2(iii) — Main Question 3 marks

A motorbike at 9 m/s accelerates uniformly at 2 m/s² for 5 s. Calculate (a) final velocity, (b) distance covered, using $s = ut + \frac{1}{2}at^2$.

Given: u = 9 m/s, a = 2 m/s², t = 5 s

(a) $v = u + at = 9 + (2 \times 5) = 9 + 10 = \mathbf{19 \text{ m/s}}$ 1 mark

(b) $s = ut + \frac{1}{2}at^2 = (9 \times 5) + \frac{1}{2}(2)(5^2) = 45 + (1 \times 25) = \mathbf{70 \text{ m}}$ 1 mark

$s = 45 + 25 = \mathbf{70 \text{ m}}$ 1 mark

OR

A ball is dropped from a 180 m tower. Taking g = 10 m/s², calculate (a) time to reach the ground, (b) velocity just before impact.

Given: h = 180 m, u = 0, g = 10 m/s²

(a) $h = ut + \frac{1}{2}gt^2 \Rightarrow 180 = 0 + \frac{1}{2}(10)t^2 \Rightarrow t^2 = 360/10 = 36$ 1 mark

$t = \sqrt{36} = \mathbf{6 \text{ s}}$ 1 mark

(b) $v = u + gt = 0 + (10 \times 6) = \mathbf{60 \text{ m/s}}$ 1 mark

(Check: $v^2 = 2gh = 2 \times 10 \times 180 = 3600 \Rightarrow v = 60 \text{ m/s. } \checkmark$)

Q.2(iv) — Main Question 3 marks

A cyclist accelerates from rest to 8 m/s in 4 s, then moves at this speed for 10 s more. What does the area under each part of the speed-time graph represent? Find the total distance covered.

Meaning of the areas: During the first 4 s (accelerating phase), the graph is a sloped line from 0 to 8 m/s, and the area under it (a triangle) represents the distance covered while accelerating. During the next 10 s (constant-speed phase), the graph is a horizontal line at 8 m/s, and the area under it (a rectangle) represents the distance covered at uniform speed. 1 mark

Distance in accelerating phase (triangle): $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 4 \times 8 = 16 \text{ m}$ 1 mark

Distance in constant-speed phase (rectangle): $8 \times 10 = 80 \text{ m}$

Total distance = $16 + 80 = 96 \text{ m}$ 1 mark

OR

Define distance and displacement. A hiker walks 5 km north, 12 km east, then 5 km south. Calculate (a) total distance, (b) net displacement.

Distance is the total length of the path travelled by a body, regardless of direction (a scalar quantity).

Displacement is the shortest straight-line distance between the initial and final positions, along with direction (a vector quantity). 1 mark

(a) Total distance = $5 + 12 + 5 = 22 \text{ km}$ 1 mark

(b) Net displacement: The 5 km north and 5 km south cancel out (she returns to her starting latitude), leaving only the 12 km eastward leg. **Net displacement = 12 km, due East** 1 mark

Q.2(v) — Main Question 3 marks

State Newton's second law ($F = ma$). A net force of 36 N acts on a 6 kg crate at rest. Calculate (a) its acceleration, (b) its velocity after 4 s.

Newton's second law: The acceleration of a body is directly proportional to the net force acting on it and inversely proportional to its mass, acting in the direction of the force: **$F = ma$** . 1 mark

(a) $a = F/m = 36/6 = 6 \text{ m/s}^2$ 1 mark

(b) $v = u + at = 0 + (6 \times 4) = 24 \text{ m/s}$ 1 mark

OR

Define inertia. Why does a standing bus passenger fall backward when the bus suddenly starts moving forward?

Inertia is the natural tendency of a body to resist any change in its state of rest or uniform motion. It depends on mass — the greater the mass, the greater the inertia. 1 mark

Explanation: When the bus is at rest, the standing passenger is also at rest. When the bus suddenly starts moving forward, the lower part of the passenger's body (in contact with the floor) is pushed forward along with the bus, but the upper part of the body tends to remain at rest due to inertia. 1 mark

As a result, relative to the accelerating bus, the passenger's upper body appears to be "left behind," making them **fall or lurch backward** unless they hold onto a support. 1 mark

Q.2(vi) — Main Question 3 marks

Define momentum and impulse. An airbag brings a 70 kg crash-test dummy from 15 m/s to rest in 0.2 s. Calculate (a) change in momentum, (b) average force exerted by the airbag.

Momentum is the product of a body's mass and velocity ($p = mv$), a vector quantity. **Impulse** is the product of the average force and the time for which it acts ($\text{Impulse} = F \times t$), and it equals the change in momentum produced. 1 mark

(a) $\Delta p = m(v - u) = 70 \times (0 - 15) = -1050 \text{ kg}\cdot\text{m/s}$ (magnitude 1050 kg·m/s) 1 mark

(b) $F = \Delta p / \Delta t = 1050 / 0.2 = 5250 \text{ N}$ 1 mark

OR

State the law of conservation of momentum. Skaters A (60 kg) and B (45 kg) push apart from rest on frictionless ice. If A moves at 3 m/s, find B's velocity and verify total momentum is zero.

Law of conservation of momentum: If no external force acts on a system, the total momentum of the system remains constant. 1 mark

Given: Initial total momentum = 0 (both at rest). After pushing apart: $m_A v_A + m_B v_B = 0 \Rightarrow (60)(3) + (45)v_B = 0$ 1 mark

$v_B = -180/45 = -4 \text{ m/s}$ (i.e. 4 m/s in the opposite direction to skater A) 1 mark

Verification: Total momentum = $(60)(3) + (45)(-4) = 180 - 180 = 0$. ✓ Momentum is conserved.

Q.2(vii) — Main Question 3 marks

Define Optics, Classical Mechanics and Quantum Mechanics, giving one real-world example of each.

Optics: The branch of physics that studies the nature and properties of light, its interaction with matter, and instruments that use or detect it. *Example:* the design of eyeglasses and camera lenses. 1 mark

Classical Mechanics: The branch of physics in which we study the motion of macroscopic (everyday-sized) objects under the action of forces. *Example:* calculating the trajectory of a thrown ball. 1 mark

Quantum Mechanics: The branch of physics in which we study the mathematical description of the motion and behaviour of microscopic particles (like electrons and atoms). *Example:* explaining the behaviour of electrons inside a semiconductor chip. 1 mark

OR

Differentiate science, technology and engineering, using smartphone development as an example of each field's role.

Science is a systematic field of knowledge concerned with accurate concepts about how the natural world behaves, established through hypotheses, experiments and observation. 1 mark

Technology is the practical application of scientific knowledge to develop material products or processes that satisfy human needs and desires. 1 mark

Engineering is the practical use of scientific and mathematical principles to create and construct structures, devices, and processes. 1 mark

Smartphone example: *Science* provided the underlying principles of physics (semiconductors, electromagnetism, optics for the camera). *Engineering* applied these principles, along with materials science, to design and build the physical device. *Technology* is the finished smartphone itself — the tool that combines communication, computation and other functions for everyday use.

Q.2(viii) — Main Question 3 marks

Differentiate like parallel forces and unlike parallel forces, with one everyday example of each.

Like parallel forces: Forces that are parallel to each other and act in the **same direction**. *Example:* two people pushing a stalled cart from behind with both hands, in the same direction. 1.5 marks

Unlike parallel forces: Forces that are parallel to each other but act in **opposite directions**. *Example:* turning a bicycle's handlebar, where one hand may push forward while the other pulls back (or applies a smaller force in the opposite sense). 1.5 marks

OR

State the principle of moments. A metre rule pivoted at 35 cm has a 6 N weight at 10 cm. Where must a 5 N weight hang to balance it?

Principle of moments: A body is in rotational equilibrium when the sum of clockwise moments about a pivot equals the sum of anticlockwise moments about the same pivot. 1 mark

Given: Pivot at 35 cm mark; 6 N weight at 10 cm mark (distance from pivot = $35 - 10 = 25$ cm); 5 N weight to be placed at distance d on the other side.

Clockwise moment = Anticlockwise moment $\Rightarrow 5 \times d = 6 \times 25$ 1 mark

$d = 150/5 = 30$ cm from the pivot, on the other side $\Rightarrow$ Position = $35 + 30 = 65$ cm mark 1 mark

Q.2(ix) — Main Question 3 marks

State Pascal's principle. How does it apply to a hydraulic wheelchair lift raised by a small pedal force?

Pascal's principle: Pressure applied to an enclosed fluid is transmitted equally and undiminished to every point of the fluid and to the walls of its container. 1 mark

Application: A hydraulic wheelchair lift has a small piston connected to a foot/hand pump and a much larger piston supporting the platform, both connected through an enclosed hydraulic fluid. 1 mark

A small force applied on the small piston creates a pressure that is transmitted, unchanged, through the fluid to the large piston. Since Pressure = Force/Area, and the large piston has a much bigger area, the same pressure produces a much larger force there — enough to lift a heavy wheelchair and passenger with only a small applied effort. 1 mark

OR

Differentiate a hydraulic lift from a hydraulic car brake system: what each multiplies, and one similarity/difference in construction.

Hydraulic lift: Multiplies **force** — a small input force on a narrow piston produces a much larger output force on a wide piston, used to lift heavy loads (e.g. a car). 1 mark

Hydraulic car brake system: Also uses Pascal's principle to multiply force, but its main purpose is to transmit the same pressure equally to all four brake pistons/calipers simultaneously, ensuring equal braking force at every wheel. 1 mark

Similarity: Both use an enclosed, incompressible hydraulic fluid and pistons of different areas, obeying Pascal's principle. **Difference:** A hydraulic lift has just two pistons (one narrow, one wide) for lifting; a brake system has one master cylinder connected via brake lines to *multiple* smaller pistons (callipers) at each wheel. 1 mark

Q.2(x) — Main Question 3 marks

Define pressure and its SI unit. A wheelbarrow wheel exerts 900 N over a contact area of 0.006 m². Calculate the pressure exerted.

Definition: Pressure is the force acting perpendicularly per unit area: $P = F/A$. **SI unit:** pascal (Pa), where 1 Pa = 1 N/m². 1 mark

Given: $F = 900 \text{ N}$, $A = 0.006 \text{ m}^2$

$P = F/A = 900/0.006$ 1 mark

$P = 150,000 \text{ Pa} = 1.5 \times 10^5 \text{ Pa}$ 1 mark

OR

Using $P = F/A$, explain why stiletto heels sink into soft ground while snow shoes do not, for similar body weight.

For a given force (body weight), pressure is **inversely proportional** to the contact area ($P = F/A$): the smaller the area, the greater the pressure, and vice versa. 1 mark

Stiletto heels: The heel has a very small contact area with the ground, so the same body weight is concentrated over a tiny area, producing very high pressure — enough to sink into soft ground. 1 mark

Snow shoes: These have a large surface area, spreading the same body weight over a much bigger area, producing low pressure that is not enough to break through the snow's surface. 1 mark

Q.2(xi) — Main Question 3 marks

Explain how biomass energy is obtained and used as fuel, naming two forms it can take.

Biomass energy comes from organic matter such as wood, crop residue, and animal waste, which stores chemical energy originally captured from sunlight. 1 mark

Form 1 — Direct combustion: Biomass (e.g. wood, rice husk) can be burned directly to produce heat, or used in electricity generation. 1 mark

Form 2 — Biofuels: Processes like fermentation, pyrolysis and gasification convert biomass into biofuels such as **ethanol** and **biodiesel**, or into syngas. 1 mark

OR

Explain how geothermal energy generates electricity, and state one advantage over fossil fuels.

Geothermal energy comes from the heat stored within the Earth's crust. Wells are drilled down to hot underground reservoirs; the heat converts water into steam, which is brought to the surface to **drive turbines connected to generators**, producing electricity. 2 marks

Advantage: Geothermal energy is abundant, reliable (available continuously, unlike solar/wind) and does not produce greenhouse gas emissions, unlike the combustion of fossil fuels. 1 mark

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3(i) — Main Question 5 marks

(a) Derive $s = ut + \frac{1}{2}at^2$ using a speed-time graph. [2] **(b) A car at 12 m/s accelerates uniformly to 24 m/s in 8 s. Find (i) acceleration, (ii) distance covered. [2]** **(c) Verify (b)(ii) using $v^2 = u^2 + 2as$. [1]**

(a) Derivation of $s = ut + \frac{1}{2}at^2$: 2 marks

Consider a body with initial velocity u accelerating uniformly at rate a , reaching velocity v after time t . On a speed-time graph, this is a straight line rising from u to v . The distance travelled s equals the area under the graph, which is a trapezium made of a rectangle ($u \times t$) and a triangle ($\frac{1}{2} \times t \times (v-u)$):

$$s = ut + \frac{1}{2}t(v - u)$$

Since $a = (v-u)/t$, we have $(v-u) = at$. Substituting:

$$s = ut + \frac{1}{2}at^2$$

(b) Numerical ($u = 12 \text{ m/s}$, $v = 24 \text{ m/s}$, $t = 8 \text{ s}$):

(i) Acceleration: $a = (v-u)/t = (24-12)/8 = 12/8 = 1.5 \text{ m/s}^2$ 1 mark

(ii) Distance: $s = ut + \frac{1}{2}at^2 = (12 \times 8) + \frac{1}{2}(1.5)(8^2) = 96 + (0.75 \times 64) = 96 + 48 = 144 \text{ m}$ 1 mark

(c) Verification using $v^2 = u^2 + 2as$: 1 mark

$$v^2 = u^2 + 2as \Rightarrow 24^2 = 12^2 + 2(1.5)(144) \Rightarrow 576 = 144 + 432 = 576. \checkmark \text{ Confirmed.}$$

OR

(a) State the three equations of motion, defining each symbol. [2] **(b) A ball is thrown upward at 18 m/s. Taking $g = 10 \text{ m/s}^2$, find the time to reach maximum height. [1]** **(c) Find the maximum height, and verify using $v^2 = u^2 - 2gh$. [2]**

(a) The three equations of motion: 2 marks

- $v = u + at$
- $s = ut + \frac{1}{2}at^2$
- $v^2 = u^2 + 2as$

where u = initial velocity, v = final velocity, a = uniform acceleration, t = time taken, s = distance covered.

(b) Time to reach maximum height ($u = 18 \text{ m/s}$, $g = 10 \text{ m/s}^2$ acting as deceleration, $v = 0$ at the top):

$$v = u - gt \Rightarrow 0 = 18 - 10t \Rightarrow t = 18/10 = 1.8 \text{ s}$$
 1 mark

(c) Maximum height: 2 marks

$$h = ut - \frac{1}{2}gt^2 = (18 \times 1.8) - \frac{1}{2}(10)(1.8^2) = 32.4 - (5 \times 3.24) = 32.4 - 16.2 = 16.2 \text{ m}$$

Verification: $v^2 = u^2 - 2gh \Rightarrow 0 = 18^2 - 2(10)(16.2) = 324 - 324 = 0. \checkmark \text{ Confirmed.}$

Q.3(ii) — Main Question 5 marks

(a) State the law of conservation of momentum for an isolated system. [1] (b) A 4 kg rifle fires a 0.02 kg bullet at 400 m/s. Find the rifle's recoil velocity. [2] (c) If the bullet is in the barrel for 0.005 s, find the average force on it, and (by Newton's third law) on the rifle. [2]

(a) Law of conservation of momentum: If no external force acts on a system of bodies, the total momentum of the system remains constant (before = after). 1 mark

(b) Recoil velocity of the rifle: 2 marks

Given: $m_{\text{rifle}} = 4 \text{ kg}$, $m_{\text{bullet}} = 0.02 \text{ kg}$, $v_{\text{bullet}} = 400 \text{ m/s}$. Before firing, total momentum = 0 (both at rest).

By conservation of momentum: $m_{\text{rifle}}v_{\text{rifle}} + m_{\text{bullet}}v_{\text{bullet}} = 0 \Rightarrow (4)v_{\text{rifle}} + (0.02)(400) = 0$

$4v_{\text{rifle}} = -8 \Rightarrow v_{\text{rifle}} = -2 \text{ m/s}$ (i.e. 2 m/s in the direction opposite to the bullet, into the shoulder)

(c) Average force on the bullet and rifle: 2 marks

Impulse on bullet = change in momentum of bullet = $m_{\text{bullet}}(v - 0) = 0.02 \times 400 = 8 \text{ kg}\cdot\text{m/s}$

Average force on bullet $F = \Delta p / \Delta t = 8 / 0.005 = 1600 \text{ N}$

By **Newton's third law**, the bullet exerts an equal and opposite reaction force on the rifle: the rifle experiences **1600 N**, directed backward (this is felt as recoil "kick" on the shoulder).

OR

(a) State how momentum conservation applies to an explosion of a stationary body into two fragments. [1] (b) A stationary 10 kg bomb explodes into a 4 kg fragment at 15 m/s and a 6 kg fragment. Find the second fragment's velocity. [2] (c) Find the total kinetic energy released. [2]

(a) Before the explosion, the total momentum of the stationary body is zero. Since no external force acts during the explosion, momentum is conserved: the total momentum of the two fragments after explosion must also equal zero, so they fly apart in exactly opposite directions with momenta of equal magnitude.

1 mark

(b) Velocity of the 6 kg fragment: 2 marks

$m_1v_1 + m_2v_2 = 0 \Rightarrow (4)(15) + (6)v_2 = 0 \Rightarrow 60 + 6v_2 = 0$

$v_2 = -60/6 = -10 \text{ m/s}$ (i.e. 10 m/s in the opposite direction to the 4 kg fragment)

(c) Total kinetic energy released: 2 marks

$\text{KE} = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}(4)(15^2) + \frac{1}{2}(6)(10^2) = \frac{1}{2}(4)(225) + \frac{1}{2}(6)(100)$

$\text{KE} = 450 + 300 = 750 \text{ J}$

Q.3(iii) — Main Question 5 marks

(a) State Pascal's principle and explain its use in a hydraulic press that compacts scrap metal. [2]

(b) A scrap-metal press has pistons of area 0.0025 m^2 and 0.6 m^2 . For an 80 N input force, find the compacting force delivered. [3]

(a) Pascal's principle: Pressure applied to an enclosed fluid is transmitted equally and undiminished to every point of the fluid and to the walls of the container. 1 mark

Application to a scrap-metal press: A small force is applied on a narrow piston, which creates pressure in the enclosed hydraulic fluid. This pressure is transmitted unchanged to a much wider piston (the ram) that presses down on the scrap metal. Because $P = F/A$ is the same on both sides, the much larger area of the ram produces a hugely magnified compacting force from a small input force. 1 mark

(b) Numerical: 3 marks

Given: A_1 (small piston) = 0.0025 m^2 , A_2 (large piston/ram) = 0.6 m^2 , $F_1 = 80 \text{ N}$

By Pascal's principle: $P_1 = P_2 \Rightarrow F_1/A_1 = F_2/A_2 \Rightarrow F_2 = F_1 \times (A_2/A_1)$

$$F_2 = 80 \times (0.6/0.0025) = 80 \times 240$$

$$F_2 = 19,200 \text{ N} = 1.92 \times 10^4 \text{ N}$$

OR

(a) Define a manometer and the principle behind its use. [2] (b) A U-tube manometer (liquid density 950 kg/m^3) shows a 32 cm height difference across a gas supply. Taking atmospheric pressure as $1.0 \times 10^5 \text{ Pa}$ and $g = 10 \text{ m/s}^2$, find (i) the pressure difference, (ii) the absolute gas pressure. [3]

(a) Manometer: A manometer is a U-shaped tube, partially filled with a liquid, used to measure the pressure of a gas. One end of the tube is connected to the gas supply, and the other end is open to the atmosphere. 1 mark

The gas pressure pushes the liquid down on the connected side and up on the open side. At equilibrium, the difference in pressure between the gas and the atmosphere equals the pressure exerted by the difference in liquid column heights (h) between the two limbs: $P_{\text{diff}} = \rho gh$. 1 mark

(b) Numerical: 3 marks

Given: $\rho = 950 \text{ kg/m}^3$, $h = 32 \text{ cm} = 0.32 \text{ m}$, atmospheric pressure = $1.0 \times 10^5 \text{ Pa}$, $g = 10 \text{ m/s}^2$

(i) $P_{\text{diff}} = \rho gh = 950 \times 10 \times 0.32 = 3040 \text{ Pa}$

(ii) Absolute gas pressure = Atmospheric pressure + $P_{\text{diff}} = 100,000 + 3040 = 103,040 \text{ Pa}$ ($1.0304 \times 10^5 \text{ Pa}$)

Q.3(iv) — Main Question 5 marks

(a) State the law of conservation of energy. [1] (b) A 6 kg object is dropped from rest from 20 m. Taking $g = 10 \text{ m/s}^2$, find (i) PE at the top, (ii) KE and speed just before impact. [4]

(a) **Law of conservation of energy:** Energy can neither be created nor destroyed; it can only be converted from one form to another, and the total energy of an isolated system remains constant. 1 mark

(b) **Numerical** ($m = 6 \text{ kg}$, $h = 20 \text{ m}$, $g = 10 \text{ m/s}^2$, $u = 0$):

(i) **PE at the top:** $E_p = mgh = 6 \times 10 \times 20 = 1200 \text{ J}$ 2 marks

(ii) **KE and speed just before impact** (no air resistance, so all PE converts to KE): 2 marks

$$E_k = E_p = 1200 \text{ J}$$

$$\frac{1}{2}mv^2 = 1200 \Rightarrow v^2 = (2 \times 1200)/6 = 400 \Rightarrow v = \sqrt{400} = 20 \text{ m/s}$$

OR

(a) Differentiate renewable and non-renewable resources, naming two renewables besides solar, wind and hydro. [2] (b) A tidal barrage drops 5000 kg of water through 8 m every second. Taking $g = 10 \text{ m/s}^2$, find (i) PE released/s, (ii) power generated. [3]

(a) 2 marks

Renewable	Non-Renewable
Replenished continuously by natural processes; will not run out.	Present in limited quantity; cannot be replaced on a human timescale.
Examples (besides solar, wind, hydro): tidal energy, geothermal energy, biomass, wave energy.	Examples: coal, oil, natural gas, nuclear (uranium) fuel.

(b) **Numerical** ($m = 5000 \text{ kg}$, $h = 8 \text{ m}$ per second, $g = 10 \text{ m/s}^2$): 3 marks

(i) PE released per second = $mgh = 5000 \times 10 \times 8 = 400,000 \text{ J} = 4 \times 10^5 \text{ J per second}$

(ii) Power = Energy/time = $400,000 \text{ J} / 1 \text{ s} = 400,000 \text{ W} = 400 \text{ kW}$ (assuming all the released PE is converted to electrical energy)